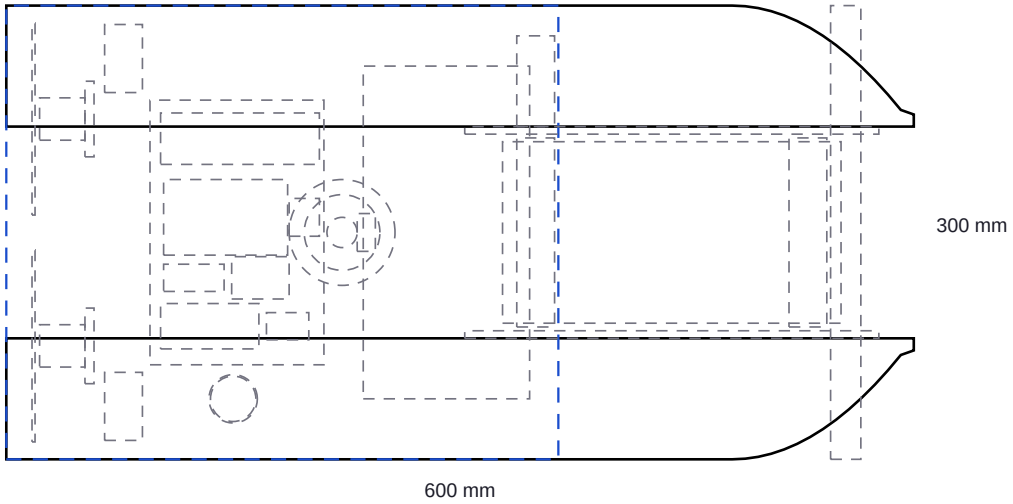
<- must measure 50 mm

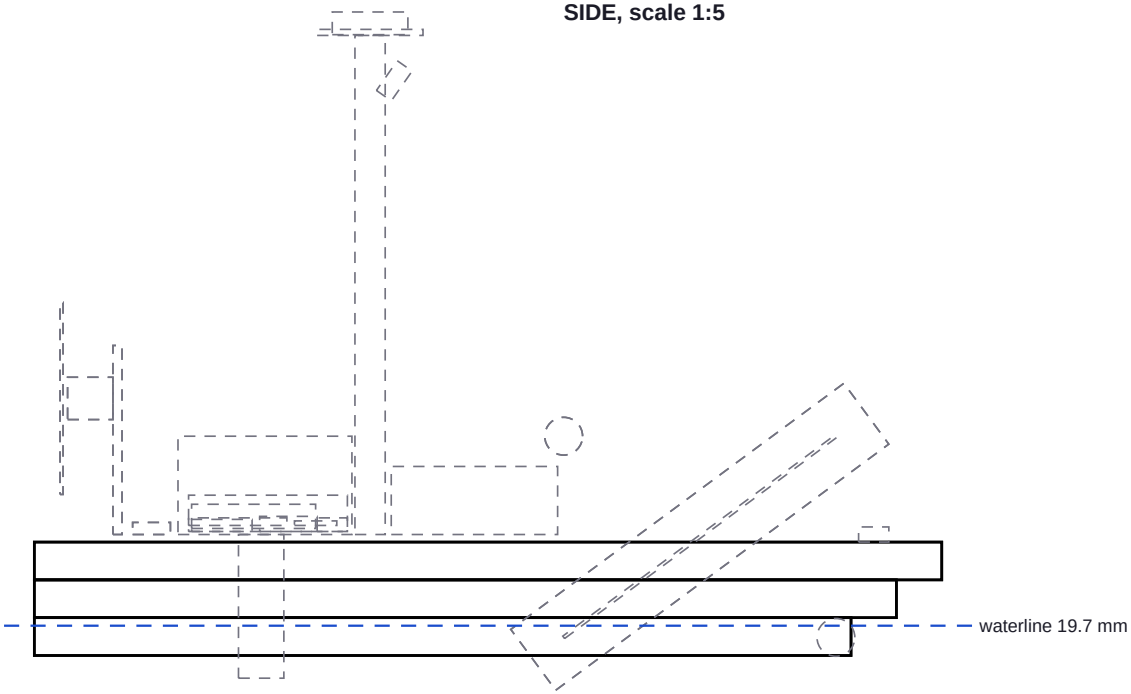
VARUNA Conveyor Cat

600 x 300 mm thermocol boat | Advanced

PLAN (from above), scale 1:5



SIDE, scale 1:5



Size	600 x 300 mm, hull 75 mm tall, 426 mm to GPS top
Weight	1563 g empty, 1781 g with 218 g of plastic
Draft	19.7 mm empty, 22.5 mm full
Reserve buoyancy	4.5 kg before the deck goes under
Stability (GM)	534 mm: very stable
Trim	bow 13.1 mm higher than stern empty; slide the battery to level
Collection	1.09 L deck bin
Camera	360 mm above water, tilted 35 deg down
Battery	LiPo 3S 2200 mAh (24 Wh), about 0.8 h
Thermocol	1 sheet 1000 x 500 x 25 mm

How to use these templates

1. Print at 100% / Actual size. Check the 50 mm bar with a ruler.
2. Join tiles by matching the grey 50 mm grid and the tile letters.
3. Stick the paper on thermocol with a glue stick.
4. Cut with a hot-wire cutter or a new snap-off blade, held upright.
5. Stack layers on skewers through the + holes. Glue with Fevicol SH.
6. Never use Fevikwik, spray paint or solvent glue: they melt thermocol.
7. Sand the bow steps smooth, coat with PVA + tissue or packing tape, then water-based acrylic paint.
8. Float-test in a tub before fitting the electronics.

cut line

cut out (window / pocket)

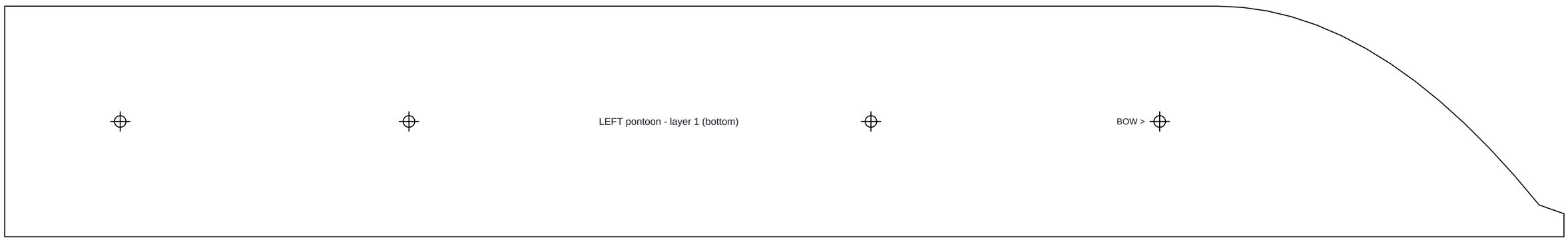
fold / glue line

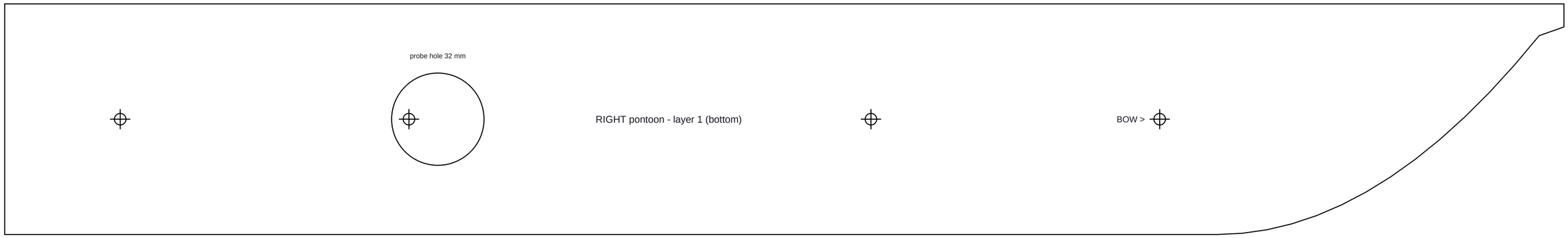
placement guide, do not cut

skewer hole: lines up the layers

VARUNA Conveyor Cat | Pontoons layer 1 (bottom), left + right | 25 mm thermocol | A2, print at 100%

Cut both. Push skewers through the dots to line up the layers.

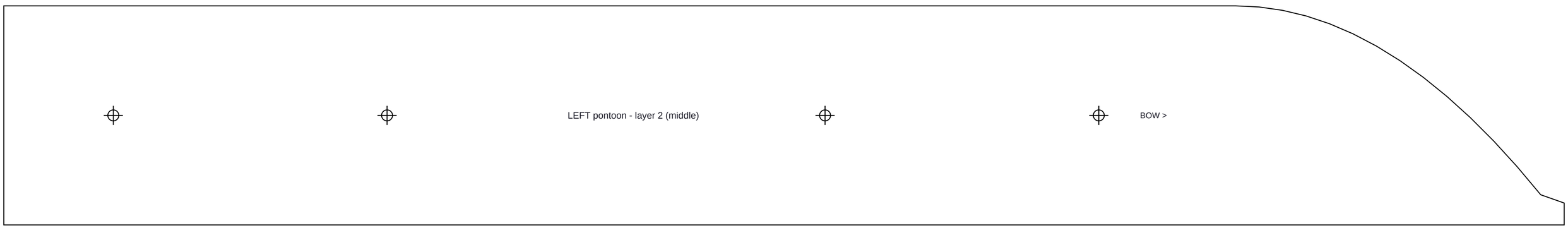


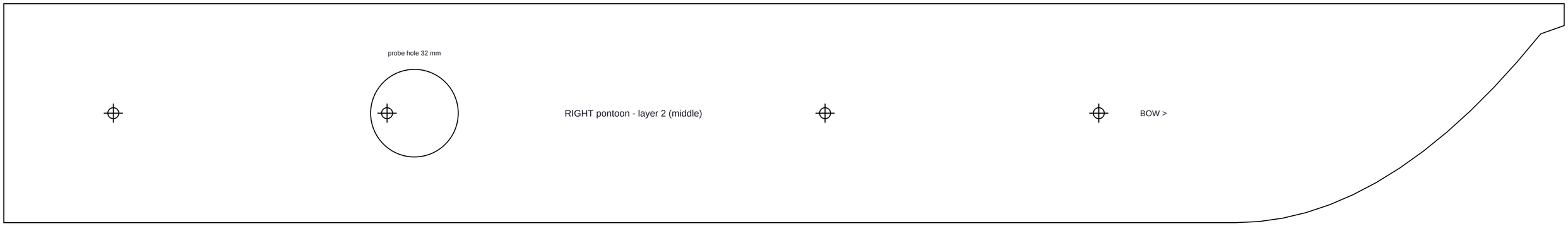


<- must measure 100 mm

VARUNA Conveyor Cat | Pontoons layer 2 (middle), left + right | 25 mm thermocol | A2, print at 100%

Cut both. Push skewers through the dots to line up the layers.

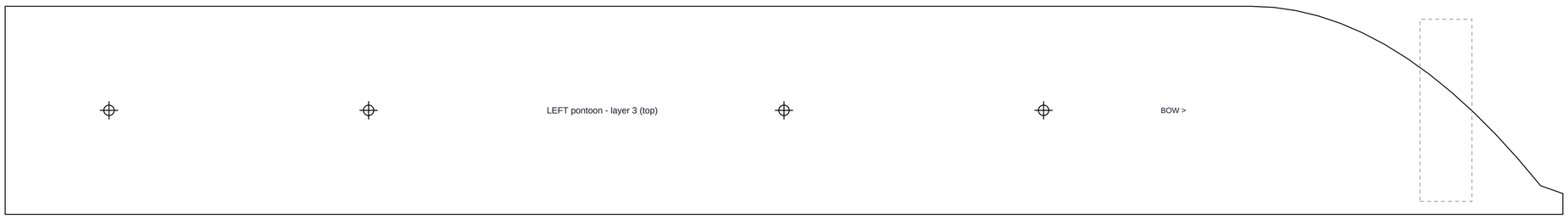


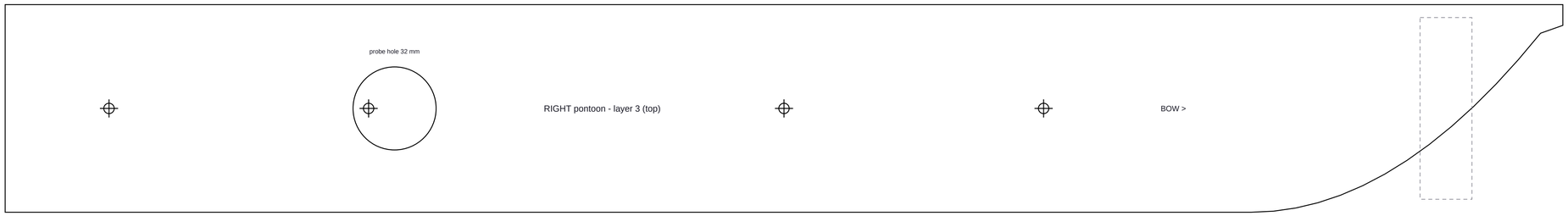


<- must measure 100 mm

VARUNA Conveyor Cat | Pontoons layer 3 (top), left + right | 25 mm thermocol | A1, print at 100%

Cut both. Push skewers through the dots to line up the layers.

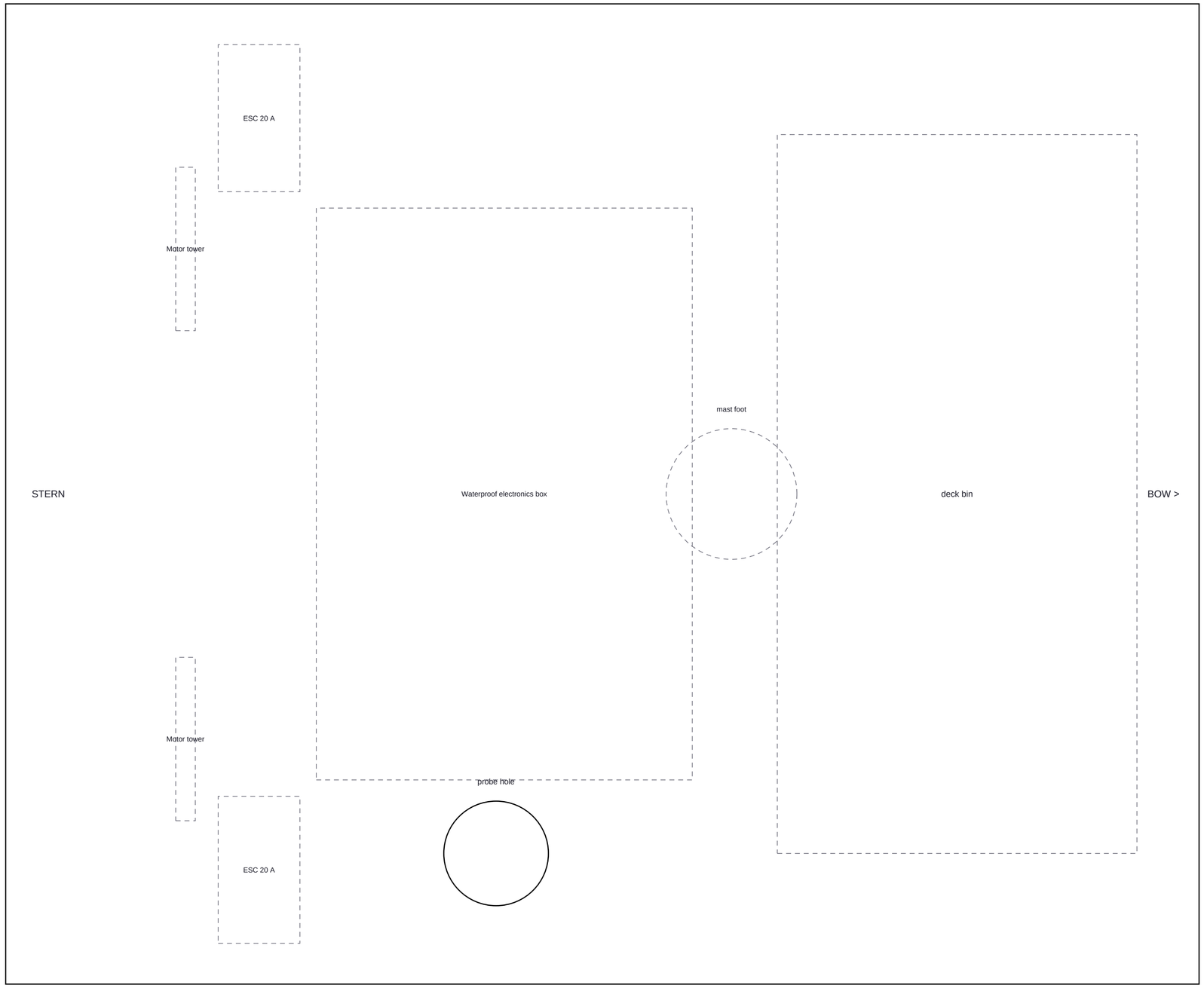




<- must measure 100 mm

VARUNA Conveyor Cat | Deck | 5 mm sunboard | A2, print at 100%

Dashed boxes show where each part is glued.

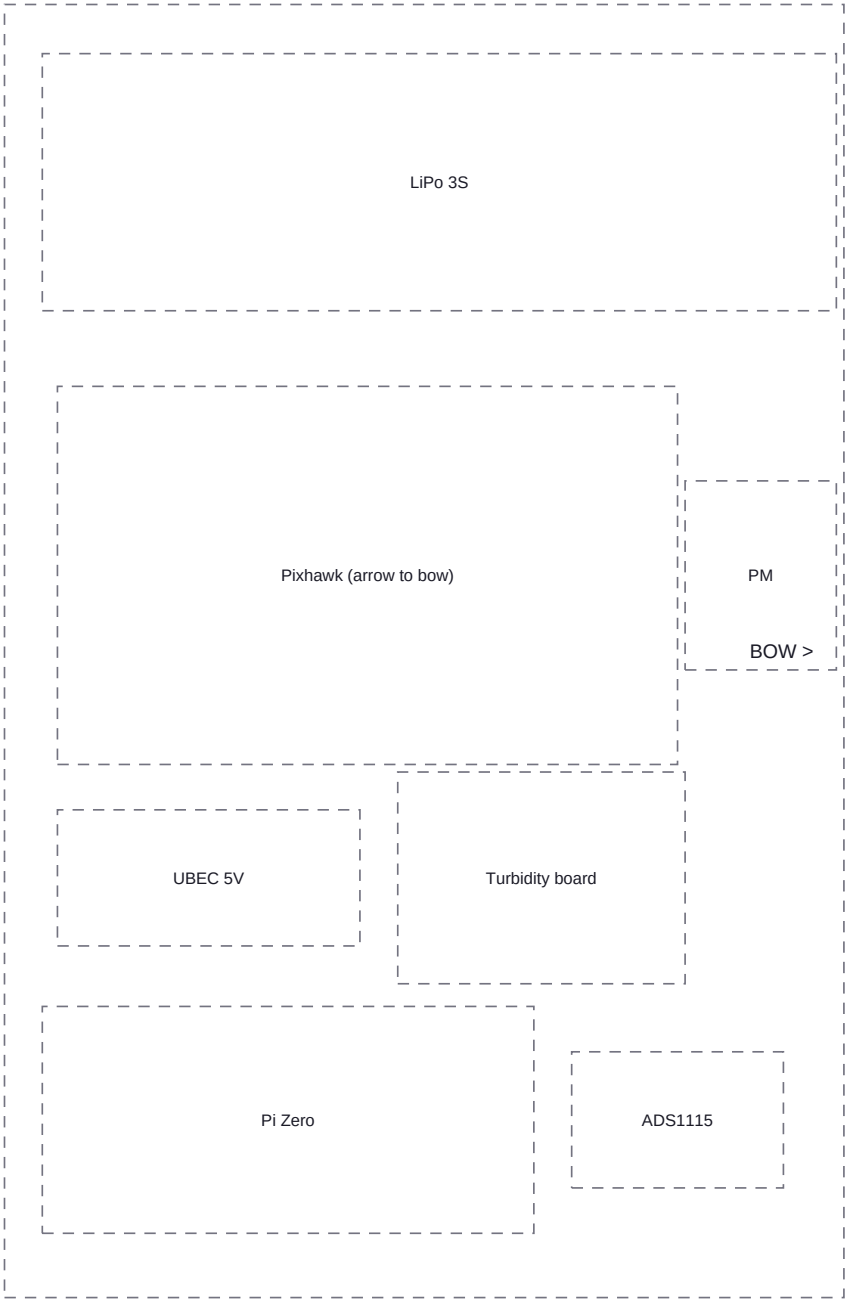


<- must measure 100 mm

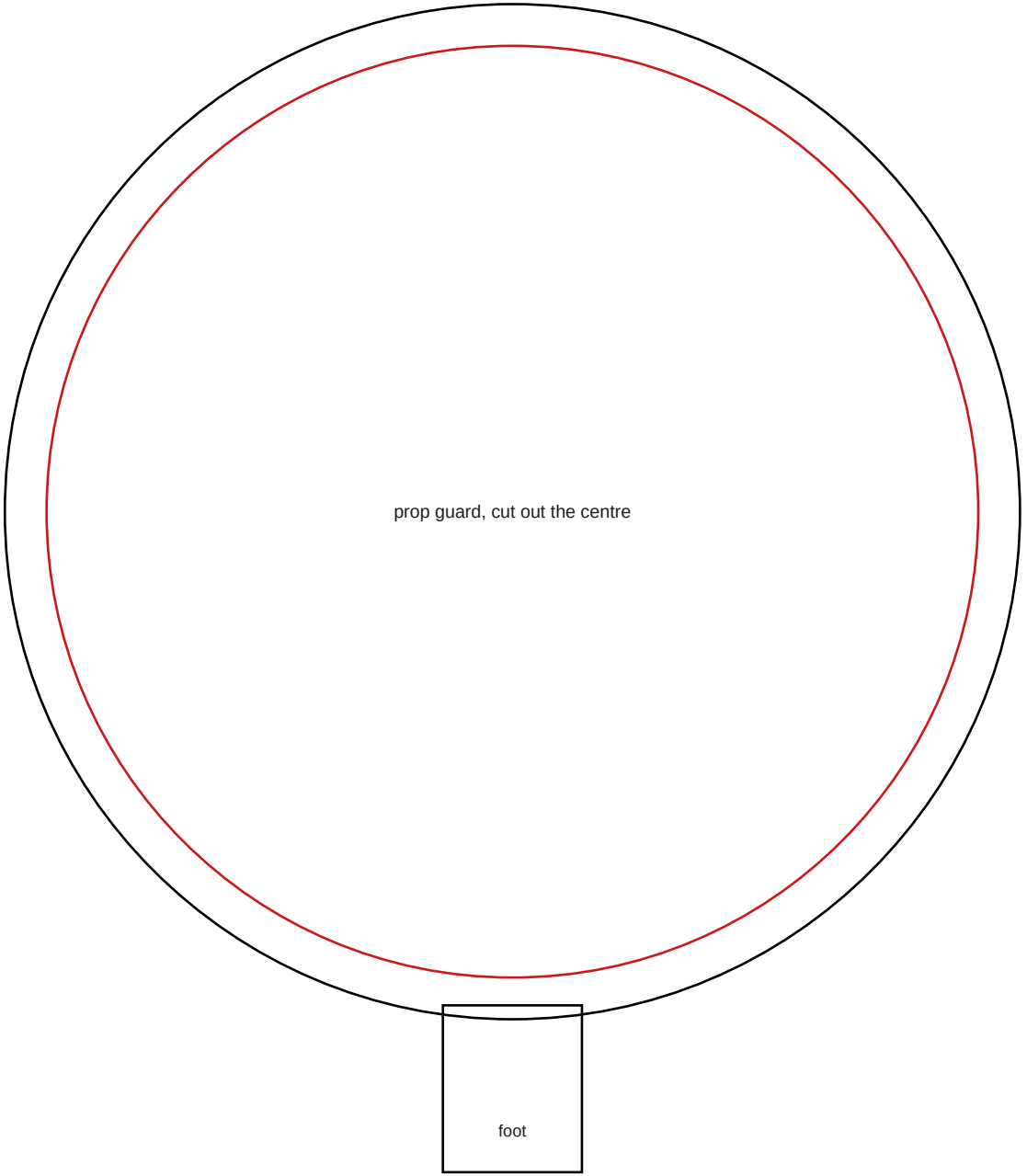
VARUNA Conveyor Cat | small parts 1 of 7

<- must measure 50 mm

Electronics box floor guide
paper guide, glue inside the floor

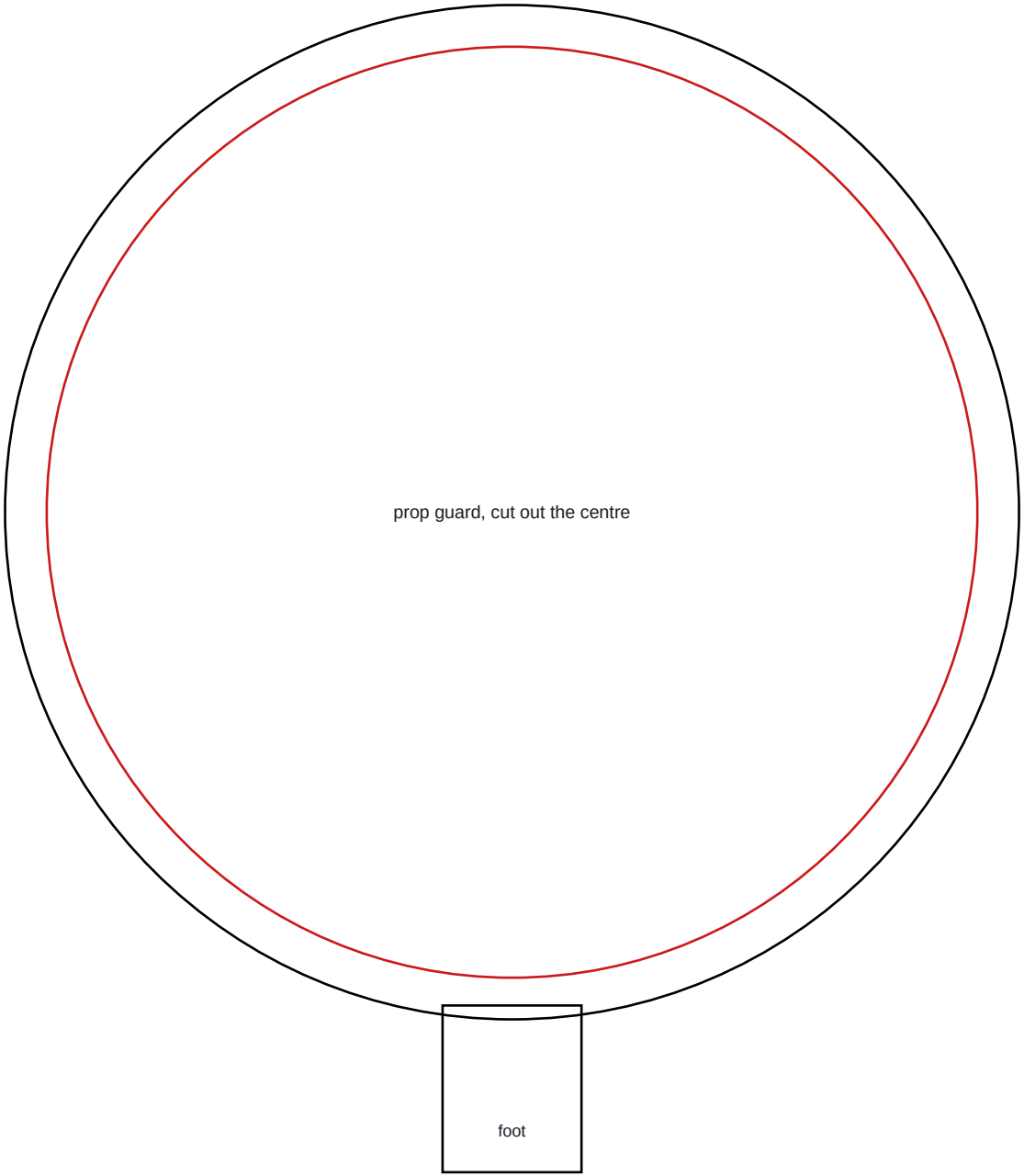


Prop guard ring (left)
5 mm sunboard



<- must measure 50 mm

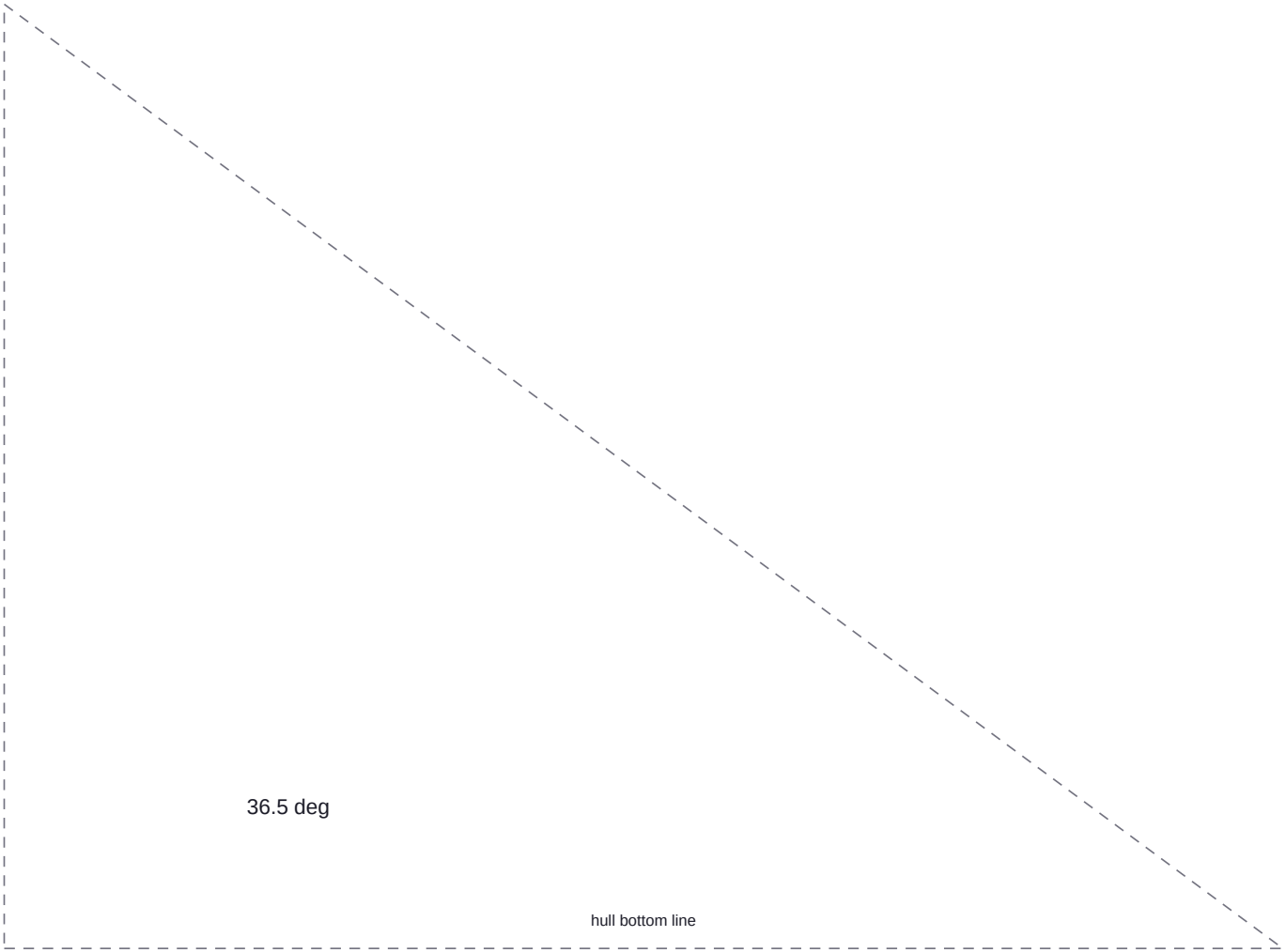
Prop guard ring (right)
5 mm sunboard



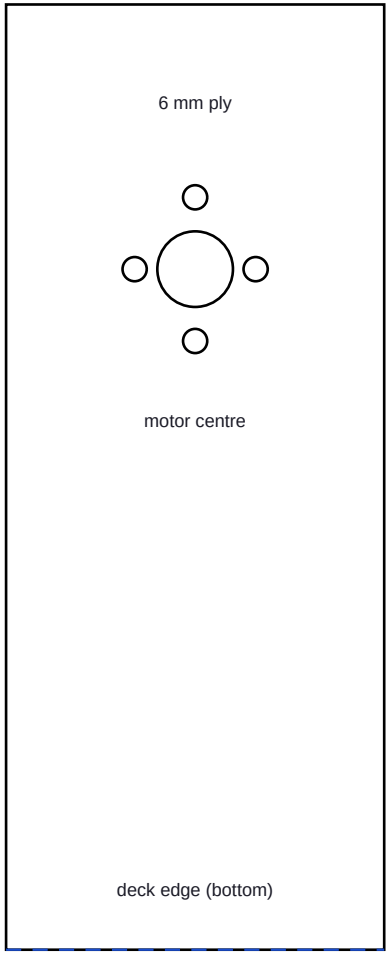
VARUNA Conveyor Cat | small parts 3 of 7

<- must measure 50 mm

Conveyor angle guide
paper or card

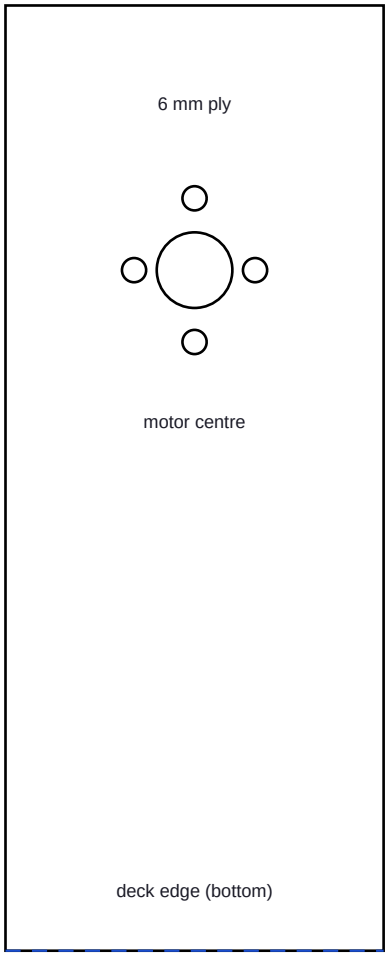


Motor tower plate (left)
6 mm plywood (or 2 x 5 mm sunboard)



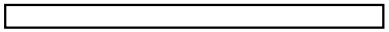
<- must measure 50 mm

Motor tower plate (right)
6 mm plywood (or 2 x 5 mm sunboard)



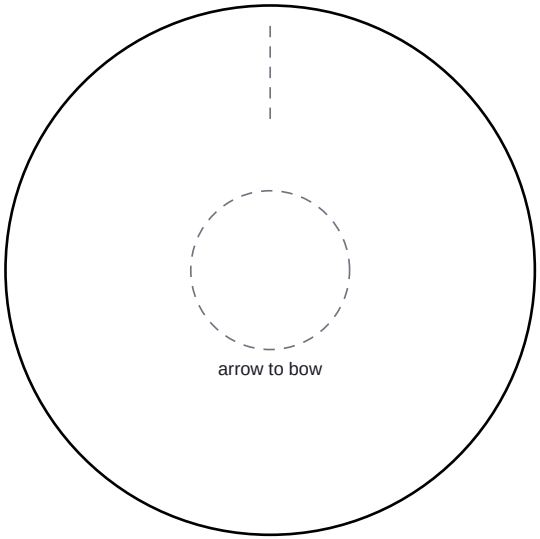
Bin bottom (drain holes)
3 mm sunboard

VARUNA Conveyor Cat | small parts 5 of 7

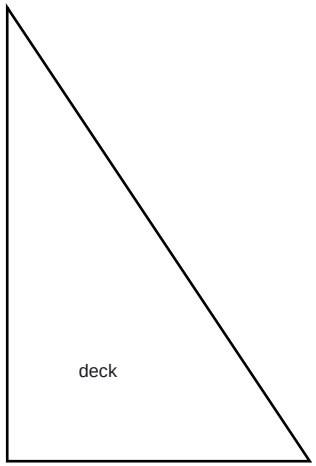


<- must measure 50 mm

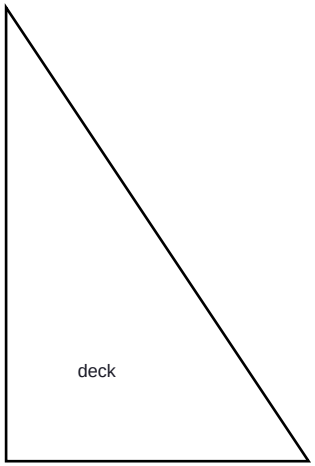
GPS disc
5 mm sunboard



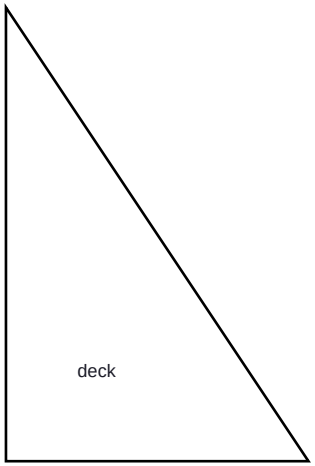
Tower gusset
5 mm sunboard



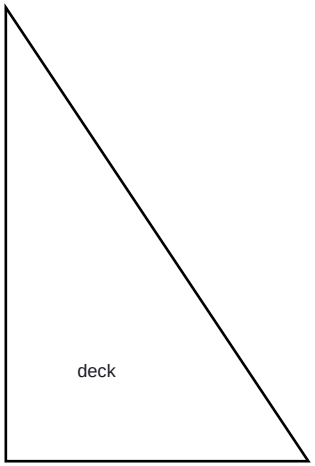
Tower gusset
5 mm sunboard



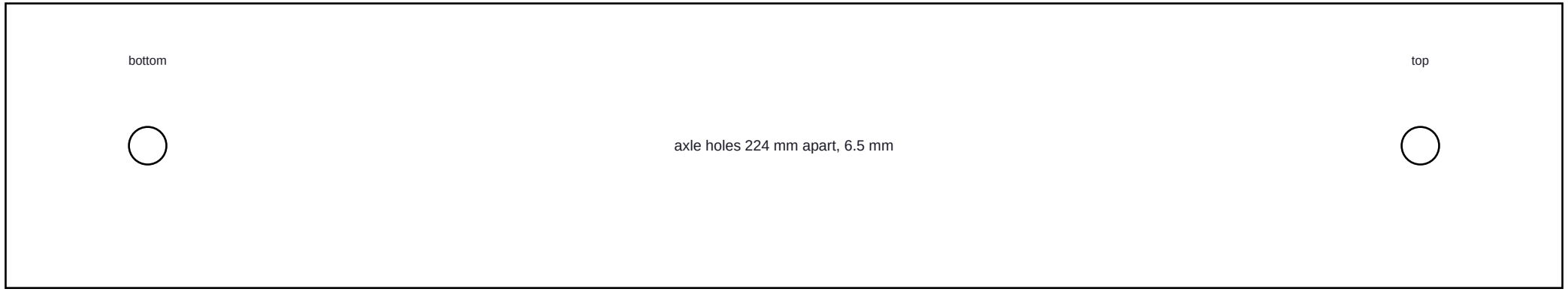
Tower gusset
5 mm sunboard



Tower gusset
5 mm sunboard

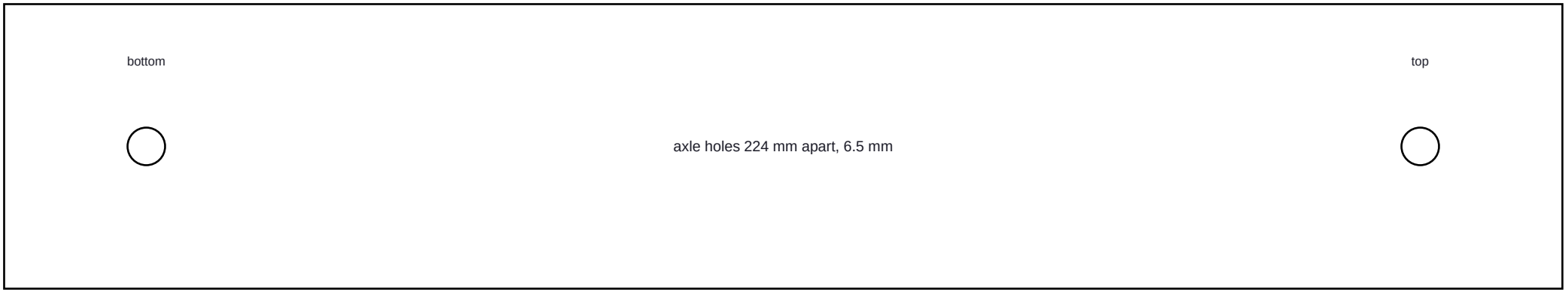


Conveyor side plate
5 mm sunboard (cut 2)

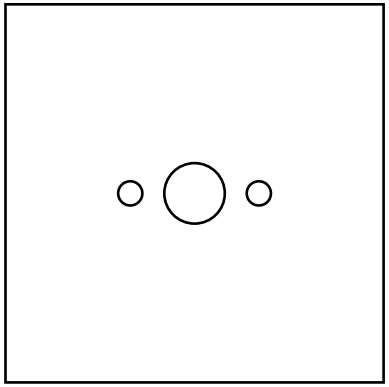


<- must measure 50 mm

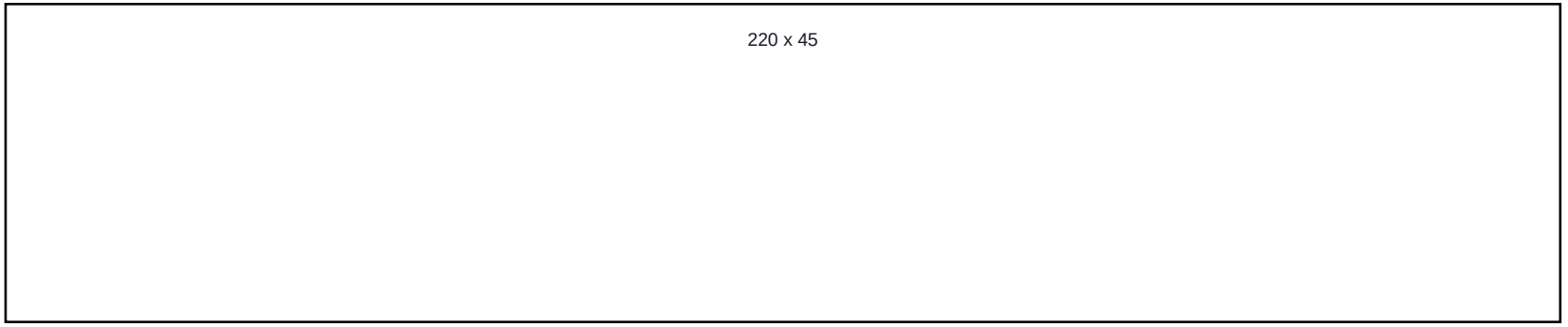
Conveyor side plate
5 mm sunboard (cut 2)



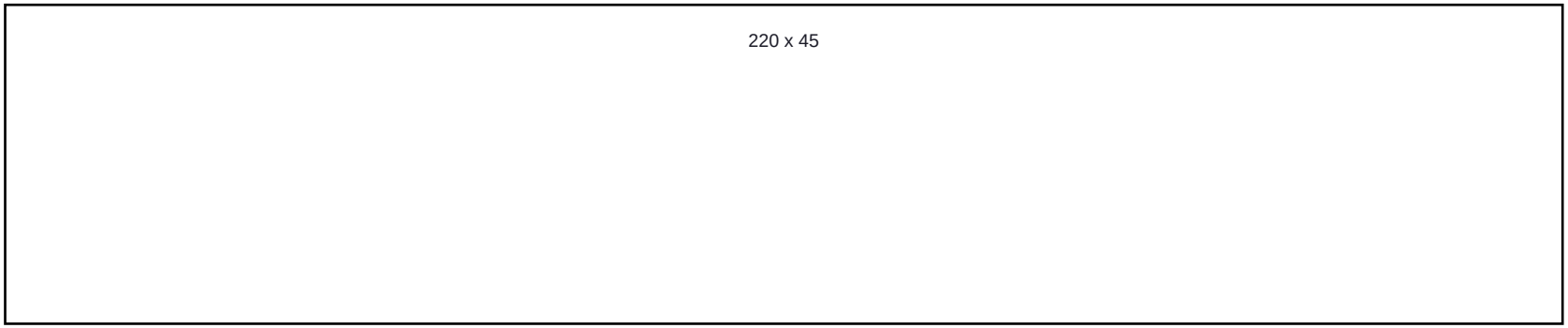
Gear motor bracket
5 mm sunboard



Bin wall front
3 mm sunboard

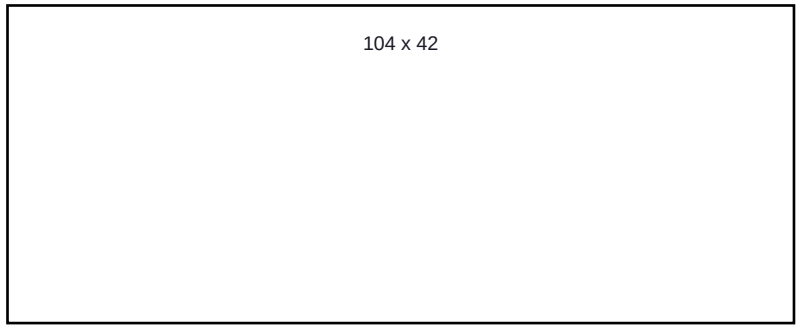


Bin wall back
3 mm sunboard

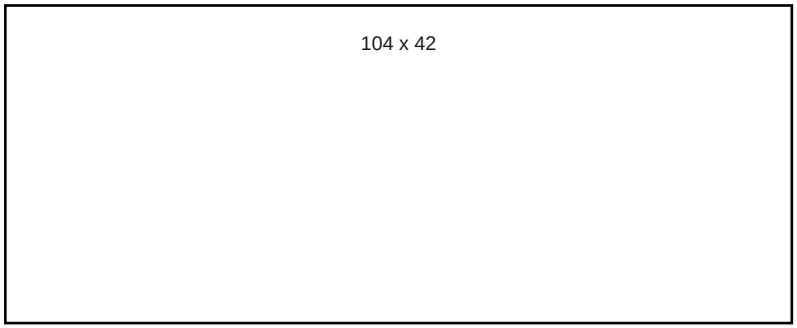


<- must measure 50 mm

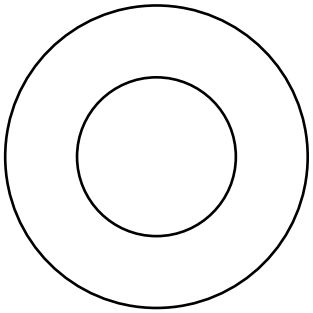
Bin wall side
3 mm sunboard



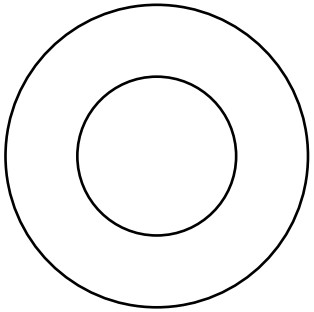
Bin wall side
3 mm sunboard



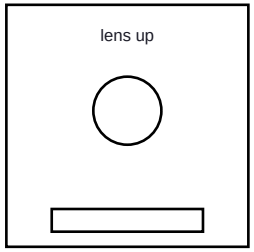
Mast foot ring
5 mm sunboard (cut 2, stack)



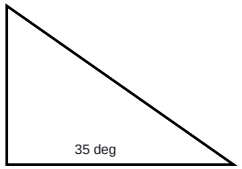
Mast foot ring
5 mm sunboard (cut 2, stack)



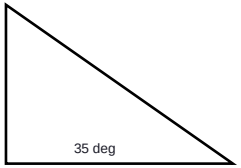
Camera plate
3 mm sunboard



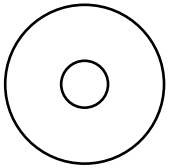
Camera wedge side (35 deg)
3 mm sunboard



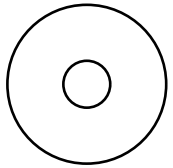
Camera wedge side (35 deg)
3 mm sunboard



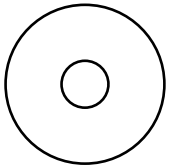
Roller end plug
5 mm sunboard (cut 8, stack 2 per end)



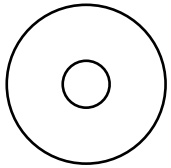
Roller end plug
5 mm sunboard (cut 8, stack 2 per end)



Roller end plug
5 mm sunboard (cut 8, stack 2 per end)



Roller end plug
5 mm sunboard (cut 8, stack 2 per end)



Belt cleat
3 mm sunboard (cut 10)

Belt cleat
3 mm sunboard (cut 10)

Belt cleat
3 mm sunboard (cut 10)

Belt cleat
3 mm sunboard (cut 10)

Belt cleat
3 mm sunboard (cut 10)